

# Experiment 03

## Lab Tasks

1. Create a repository 'internship-diary' with a README describing yourself. Make at least 5 meaningful commits (one per section: intro, skills, goals, projects, contact). Push to GitHub and submit the link.
2. Practise the full clone workflow: clone <https://github.com/octocat/Hello-World>, explore its history with git log, and paste the oldest commit's message and date.
3. Create two branches feature-a and feature-b from main, add a different file in each, merge both into main, and screenshot the final git log --oneline --graph.
4. Write a proper .gitignore for a Node.js project (research: what should Node projects ignore?) and explain each line in comments.
5. Cause a merge conflict on purpose (edit the SAME line of README on two branches, then merge), resolve it, and document the steps with screenshots.

1. Create a repository 'internship-diary' with a README describing yourself. Make at least 5 meaningful commits (one per section: intro, skills, goals, projects, contact). Push to GitHub and submit the link.

The screenshot shows a GitHub repository page for 'ahmarhayat / internship-diary'. The commit being viewed is '112ca45', titled 'docs: add goals section to README', committed 2 minutes ago. The commit message is 'docs: add goals section to README'. The commit is on the 'main' branch and has one parent commit '7a23dd4'. The diff shows changes to 'README.md', with 4 lines added (indicated by green '+' signs). The changes include a new '## Goals' section with two bullet points: '- Master full-stack web development' and '- Learn cloud platform deployment'. The left sidebar shows the file 'README.md' selected. The bottom of the page shows 'Comments 0' and a 'Lock conversation' button.

```
MINGW64:/f/Intenship/internship-diary
Computer Arena@Hayat-PC MINGW64 /f/Intenship/internship-diary (main)
$ git log --oneline
112ca45 (HEAD -> main) docs: add goals section to README
7a23dd4 docs: add contact section to README
158674e docs: add projects section to README
4005807 docs: add skills section to README
cf380be docs: add intro section to README

Computer Arena@Hayat-PC MINGW64 /f/Intenship/internship-diary (main)
$ git remote add origin https://github.com/ahmarhayat/internship-diary.git
git push -u origin main
Enumerating objects: 15, done.
Counting objects: 100% (15/15), done.
Delta compression using up to 4 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (15/15), 1.45 KiB | 371.00 KiB/s, done.
Total 15 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), done.
To https://github.com/ahmarhayat/internship-diary.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

Computer Arena@Hayat-PC MINGW64 /f/Intenship/internship-diary (main)
$
```

2. Practice the full clone workflow: clone <https://github.com/octocat/Hello-World>, explore its history with git log, and paste the oldest commit's message and date.

```
MINGW64:/f/Intenship/Hello-World
Computer Arena@Hayat-PC MINGW64 /f/Intenship
$ cd /f/Intenship
git clone https://github.com/octocat/Hello-World.git
cd Hello-World
git log --reverse
Cloning into 'Hello-World'...
remote: Enumerating objects: 13, done.
remote: Total 13 (delta 0), reused 0 (delta 0), pack-reused 13 (from 1)
Receiving objects: 100% (13/13), done.
commit 553c2077f0edc3d5dc5d17262f6aa498e69d6f8e
Author: cameronmcefee <cameron@github.com>
Date:   Wed Jan 26 11:06:08 2011 -0800

    first commit

commit 762941318ee16e59dabbac1b4049eec22f0d303
Author: Johnneylee Jack Rollins <Johnneylee.rollins@gmail.com>
Date:   Tue Sep 13 21:42:41 2011 -0700

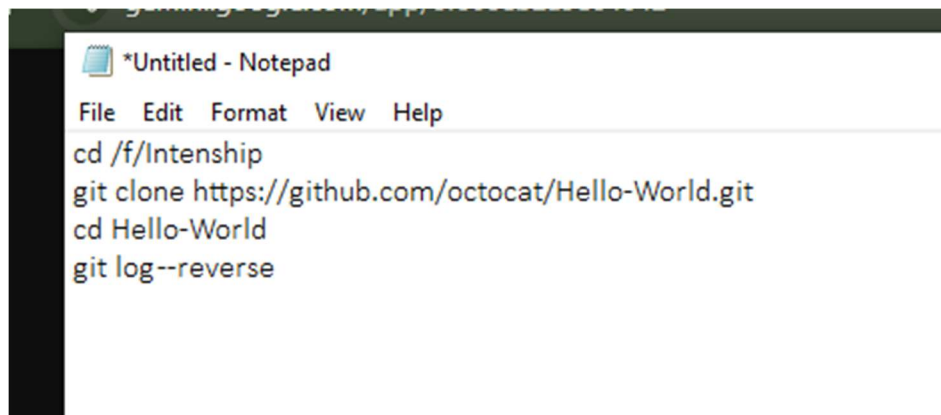
    New line at end of file. --Signed off by Spaceghost

commit 7fd1a60b01f91b314f59955a4e4d4e80d8edf11d (HEAD -> master, origin/master, origin/HEAD)
Merge: 553c207 7629413
Author: The Octocat <octocat@nowhere.com>
Date:   Tue Mar 6 15:06:50 2012 -0800

    Merge pull request #6 from Spaceghost/patch-1

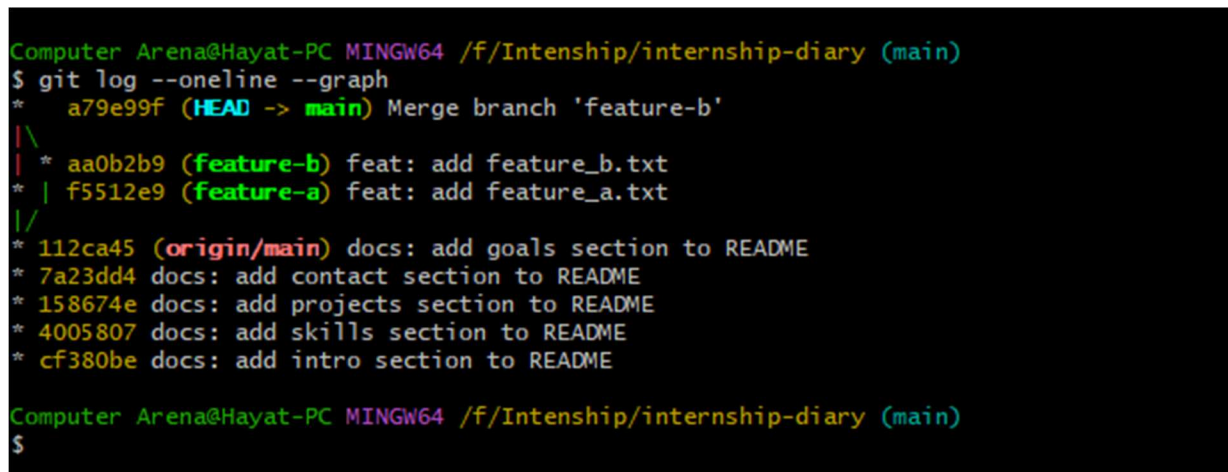
    New line at end of file.

Computer Arena@Hayat-PC MINGW64 /f/Intenship/Hello-World (master)
$
```

A screenshot of a Notepad window titled '\*Untitled - Notepad'. The window has a menu bar with 'File', 'Edit', 'Format', 'View', and 'Help'. The text inside the Notepad is a series of terminal commands:

```
cd /f/Intenship
git clone https://github.com/octocat/Hello-World.git
cd Hello-World
git log--reverse
```

3. Create two branches feature-a and feature-b from main, add a different file in each, merge both into main, and screenshot the final git log --oneline --graph.

A screenshot of a terminal window with a black background and green text. The prompt is 'Computer Arena@Hayat-PC MINGW64 /f/Intenship/internship-diary (main)'. The command '\$ git log --oneline --graph' has been executed, resulting in the following output:

```
$ git log --oneline --graph
*   a79e99f (HEAD -> main) Merge branch 'feature-b'
| \
| * aa0b2b9 (feature-b) feat: add feature_b.txt
| * f5512e9 (feature-a) feat: add feature_a.txt
| /
* 112ca45 (origin/main) docs: add goals section to README
* 7a23dd4 docs: add contact section to README
* 158674e docs: add projects section to README
* 4005807 docs: add skills section to README
* cf380be docs: add intro section to README

Computer Arena@Hayat-PC MINGW64 /f/Intenship/internship-diary (main)
$
```

4. Write a proper .gitignore for a Node.js project (research: what should Node projects ignore?) and explain each line in comments.

```

MINGW64:/f/Intenship/internship-diary

Computer Arena@Hayat-PC MINGW64 /f/Intenship/internship-diary (main)
$ cat << 'EOF' > .gitignore
> node_modules/
> .env
> .env.local
> .env.production
> /dist
> /build
> *.log
npm-debug.log*
> /coverage
> .vscode/
> .DS_Store
> Thumbs.db
EOF

Computer Arena@Hayat-PC MINGW64 /f/Intenship/internship-diary (main)
$ git add .gitignore
git commit -m "chore: add .gitignore for Node.js project"
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the ne
xt time Git touches it
[main 2e799d5] chore: add .gitignore for Node.js project
1 file changed, 12 insertions(+)
create mode 100644 .gitignore

Computer Arena@Hayat-PC MINGW64 /f/Intenship/internship-diary (main)
$ |

```

5. Cause a merge conflict on purpose (edit the SAME line of README on two branches, then merge), resolve it, and document the steps with screenshots.

```

Computer Arena@Hayat-PC MINGW64 /f/Intenship/internship-diary (main|MERGING)
$ cat conflict_test.txt
<<<<<<< HEAD
Line modified by conflict-1 branch
=====
Line modified by conflict-2 branch
>>>>>>> conflict-2

```

```

Computer Arena@Hayat-PC MINGW64 /f/Intenship/internship-diary (main)
$ git merge conflict-2
Auto-merging conflict_test.txt
CONFLICT (add/add): Merge conflict in conflict_test.txt
Automatic merge failed; fix conflicts and then commit the result.

Computer Arena@Hayat-PC MINGW64 /f/Intenship/internship-diary (main|MERGING)
$

```

```
Computer Arena@Hayat-PC MINGW64 /f/Intenship/internship-diary (main|MERGING)
$ echo "Line modified by conflict-1 and conflict-2 merged successfully" > conflict_test.txt

Computer Arena@Hayat-PC MINGW64 /f/Intenship/internship-diary (main|MERGING)
$ git add conflict_test.txt
git commit -m "fix: resolve merge conflict between conflict-1 and conflict-2"
warning: in the working copy of 'conflict_test.txt', LF will be replaced by CRLF the next time Git touches it
[main f1254e3] fix: resolve merge conflict between conflict-1 and conflict-2

Computer Arena@Hayat-PC MINGW64 /f/Intenship/internship-diary (main)
$ git push origin main
Enumerating objects: 21, done.
Counting objects: 100% (21/21), done.
Delta compression using up to 4 threads
Compressing objects: 100% (16/16), done.
Writing objects: 100% (20/20), 1.70 KiB | 116.00 KiB/s, done.
Total 20 (delta 8), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (8/8), done.
To https://github.com/ahmarhayat/internship-diary.git
   112ca45..f1254e3  main -> main

Computer Arena@Hayat-PC MINGW64 /f/Intenship/internship-diary (main)
$ |
```

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